BOLLAPALLY BANSI DEEPAK

AI Engineer / Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer and AI Engineer specializing in building end-to-end web applications and integrating AI/ML and **Computer Vision** systems. Proficient in **React**, **Next.js**, **Node.js**, and **TypeScript** for front-end and back-end development, with deep expertise in **LLM frameworks**, **LangChain**, **RAG Architectures**, and **vector search**. Experienced in deploying production-grade AI platforms using **FastAPI**, **Celery**, and **microservices**, with asynchronous processing and real-time inference at scale.

PROFESSIONAL EXPERIENCE

AI Engineer

Neurom Innovations

Dec 2024 – Present | Hyderabad, India

- Architected end-to-end RAG systems using schema-aware retrieval, enabling high-accuracy LLM responses for enterprise data sources.
- Built a dynamic multi-model inference gateway in **FastAPI** that intelligently routes traffic between **Google Gemini** and **OpenAI** based on task complexity, latency budget, and cost-efficiency.
- Built production **computer-vision** pipelines — real-time face detection, deep face-embedding models, and **pgvector** similarity search — powering a multi-tenant biometric attendance platform with **AES-256-GCM**-encrypted PII.
- Engineered real-time **geospatial** and **signal-processing** systems — live **WebGL** satellite-orbit tracking and **SDR** pipelines (FFT, FM demodulation) — on **event-driven** Node.js/TypeScript microservices.
- Reduced hallucinations by designing enterprise-grade guardrails, including policy-driven validation, tool-calling restrictions, and structured output enforcement.
- Led full-stack development of internal AI pilots, including React/Next.js dashboards, asynchronous inference queues, and real-time task monitoring.
- Delivered full-stack web applications across multiple client projects — building React/Next.js frontends, FastAPI/Express backends, and integrating AI systems end-to-end.
- Optimized LLM performance pipelines, benchmarking inference cost, throughput, caching layers, and implementing structured prompting frameworks.

KEY PROJECTS

CORAX – AI Face-Recognition Attendance Platform *(In Progress)*

FastAPI, PostgreSQL/pgvector, Celery, InsightFace, MediaPipe, React, React Native, Docker

- Architecting a multi-tenant face-recognition attendance platform end-to-end — an async **FastAPI** backend (**81 REST endpoints**, 15 data models), a **React + TypeScript** dashboard, and a **React Native** mobile app — containerized as a **7-service Docker** stack (Postgres/pgvector, Redis, Celery, nginx).
- Engineered a real-time face-recognition pipeline pairing client-side **MediaPipe** 468-landmark detection (7 Hz) with a server-side two-stage stack — an **InsightFace SCRFD** detector plus a **512-dim deep face-embedding model** — and **pgvector** cosine similarity for sub-second identity matching.
- Built an asynchronous video-processing pipeline on **Celery + Redis** that returns **HTTP 202** instantly and runs frame sampling and inference off the request path, keeping the API responsive under heavy media load.
- Implemented production-grade security — **AES-256-GCM** encryption of PII/biometric data, **JWT** auth with refresh-token rotation and revocation, rate limiting, account lockout, and JWT-enforced tenant isolation.
- Developed a data-rich **React + TypeScript** dashboard with **React Query** (90+ data hooks), **Zustand**, and **Recharts** — featuring drill-down navigation, live attendance status, and CSV/PDF reporting.
- Delivered a cross-platform **React Native (Expo)** app for on-site capture — native-camera face enrollment, offline-aware sync, biometric login, and push notifications.

ASKENCE – Enterprise Text-to-SQL & Analytics Platform

React 19, TypeScript, FastAPI, LangChain, Gemini, PostgreSQL, Docker

- Designed an advanced Text-to-SQL RAG system using **LangChain** and **Schema Injection** to generate accurate and executable SQL queries from natural language.
- Built a **SQL Validation & Rewriting Engine** that detects large datasets and automatically produces optimized, aggregated queries.
- Developed a dynamic visualization engine using **React + TypeScript + Vite** that classifies output types and renders appropriate visualizations.

TECHNICAL SKILLS

React.js · Next.js · React Native · TypeScript · JavaScript · Vite · Tailwind CSS · Cesium · WebGL

Node.js · Express · FastAPI · Python · Celery · Redis · Streamlit

Google Gemini · OpenAI API · LangChain · RAG Pipelines · Computer Vision

Vector Search · Prompt Engineering

Text-to-SQL

Pandas · NumPy · Plotly · SDR/DSP

PostgreSQL · pgvector · Docker · Kubernetes · AWS Amplify · AWS EC2 · Git · CI/CD

CERTIFICATIONS

1. Oracle Cloud Infrastructure 2023 Certified Associate
2. Postman API Fundamentals Student Expert

EDUCATION

B.Tech, Computer Science & Engineering

Talla Padmavathi College of Engineering

2021 – 2025

- Deployed system for **real-world enterprise** data analytics, enabling business users to query operational data through natural language.

ISVT GROUND STATION – Real-Time Satellite Tracking & SDR Platform *(In Progress)*

React, TypeScript, Node/Express, PostgreSQL, Redis, Cesium, satellite.js, Python, RTL-SDR

- Building a live LEO-satellite visualizer that renders real-time orbits via client-side **SGP4** propagation (**satellite.js**) in a **Cesium/WebGL** 3D scene, paired with a 2D **Mapbox** ground track and sub-second position updates.
- Engineered an **event-driven** Node/Express + **TypeScript** backend with a central **EventBus** and pluggable **connectors** that ingest TLE feeds from **CelesTrak** and **Space-Track** into **PostgreSQL** with **Redis** caching and a 2-hour refresh cron.
- Developed a **Python SDR** signal pipeline (**RTL-SDR**) performing **FM demodulation**, **FFT** spectrum analysis, and live **waterfall** rendering streamed to the React frontend in real time.
- Architected a **TypeScript monorepo** (npm workspaces) with a shared type ontology, raw-SQL migrations, and a **versioned REST API** (strict-mode, consistent response envelope); implemented orbital math — pass prediction, **Doppler** shift, and topocentric elevation/azimuth across ECI/ECEF/geodetic frames.

SEARCHLIFT360 – Asynchronous AI SEO Auditing Engine

FastAPI, Python, Docker, Next.js 14, AWS Amplify

- Architected a scalable asynchronous crawling engine using **FastAPI Background Tasks**, enabling parallel deep-site audits with zero UI blocking.
- Developed a unique hybrid analysis pipeline combining **rule-based DOM Checks** with **generative AI insights** for content strategy.
- Automated the creation of client-ready PDF audit reports using **ReportLab**, generating structured, visual SEO health summaries.